

Case Study : Smart Energy Consumption Analytics System using Apache Spark

A national electricity distribution company collects smart meter data, energy consumption records, weather information, and grid performance data. The organization wants to build an **Energy Analytics Platform** to optimize power distribution and predict energy demand.

As a Big Data Engineer, design, implement, and deploy the solution using Apache Spark and PySpark.

Q1. Spark Initialization and Data Loading (2 Marks)

Initialize Spark and load energy consumption datasets.

Q2. RDD Implementation (3 Marks)

Create RDDs and perform transformation and action operations.

Q3. Key-Value Operations and Persistence (2 Marks)

Implement key-value operations, shuffle operations, and persistence techniques.

Q4. Spark DataFrame Operations (3 Marks)

Perform DataFrame selection, filtering, grouping, joins, and aggregation.

Q5. Exploratory Data Analysis and Spark SQL (5 Marks)

Perform EDA and execute Spark SQL queries to:

- Analyze hourly energy consumption.
- Determine region-wise electricity usage.
- Identify peak demand periods.
- Analyze consumer categories.
- Generate monthly consumption reports.

Q6. ETL Pipeline Development (3 Marks)

Develop ETL pipelines for smart meter and weather datasets.

Q7. Machine Learning/Deep Learning Implementation (2 Marks)

Implement a model to predict electricity demand or detect anomalies.

Additional Implementation Instructions

A. Version Control using Git and GitHub

- Create a **GitHub repository** for the project.
- Upload all source code, datasets (or dataset links), notebooks, and documentation.
- Maintain proper folder structure and commit history.
- Include a **README.md** file containing project description, installation steps, execution procedure, and output screenshots.

B. Containerization using Docker

- Create a **Dockerfile** for the Spark application.
- Build and test the Docker image.

C. Deployment using Kubernetes

- Create Kubernetes deployment and service YAML files.
- Deploy the application on a Kubernetes cluster (Minikube/Kubernetes/OpenShift).
- Verify successful deployment.

D. DevOps Pipeline Implementation

- Configure a basic CI/CD pipeline using GitHub Actions/Jenkins/GitLab CI.
- Automate build, testing, Docker image creation, and deployment.

E. Project Submission Requirements

Submit:

- Source code
- GitHub repository link
- Dockerfile
- Kubernetes YAML files
- CI/CD pipeline configuration
- Execution screenshots
- EDA and Spark SQL outputs
- Model evaluation report
- Final project documentation